Extensions on classical E_r-pages, Generalized Leibniz Rule and Mahowald Trick

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1 Extensions on a classical E_r-page

To state the Generalized Leibniz Rule in terms of the classical Adams spectral sequence, we need to define f-extensions not only on homotopy groups but *on the E_r-page* as well.

1.1 synthetic lifts

Notation 1. *For a map between classical spectra $f : X \to Y$, consider the associated synthetic map from Notation* **??**

$$\hat{f} : \Sigma^{0,e(f)}\nu X \to \nu Y.$$

Recall that $\lambda^{e(f)}\hat{f} = \nu f$

For any $2 \leq r \leq \infty$, we denote the following mod λ^{r-1} reduction maps by $\hat{f}_{r-1}$:

$$\hat{f}_{r-1} : \Sigma^{0,e(f)}\nu X/\lambda^{r-1} \to \nu Y/\lambda^{r-1}.$$

1.1.0.1

The E_0-page of the $\hat{f}_{r-1}$-ESS is isomorphic to

$$\begin{aligned} {}^{\hat{f}_{r-1}}E_0^{s,t,t-k+e(f)} \cong & E_\infty^{s,t,t-k}(\nu X/\lambda^{r-1}) \oplus E_\infty^{s,t,t-k+e(f)}(\nu Y/\lambda^{r-1}) \\ \cong & \left(Z_{r-1-k}^{s,t}(X)/B_{1+k}^{s,t}(X)\right) \oplus \left(Z_{r-1-k+e(f)}^{s,t}(Y)/B_{1+k-e(f)}^{s,t}(Y)\right). \end{aligned}$$

1.1.0.2

A nontrivial $d_n^{\hat{f}_{r-1}}$ differential can be interpreted as a map from the subgroup

$$ {}^{\hat{f}_{r-1}}Z_{n-1}^{s,t,t-k}(X) \subset Z_{r-1-k}^{s,t}(X)/B_{1+k}^{s,t}(X) \tag{2} $$

to the quotient group

$$ \left(Z_{r-1-k-n+e(f)}^{s+n,t+n}(Y)/B_{1+k+n-e(f)}^{s+n,t+n}(Y)\right)/\,{}^{\hat{f}_{r-1}}B_{n-1}^{s+n,t+n,t-k}(Y). \tag{3} $$

1.1.0.3

The differential $d_n^{\hat{f}_{r-1}}$ is trivial for degree reasons when

$$ n < \mathrm{AF}(f) \text{ or } n > r-2-k+e(f). $$

1.2 extension on the E_r page

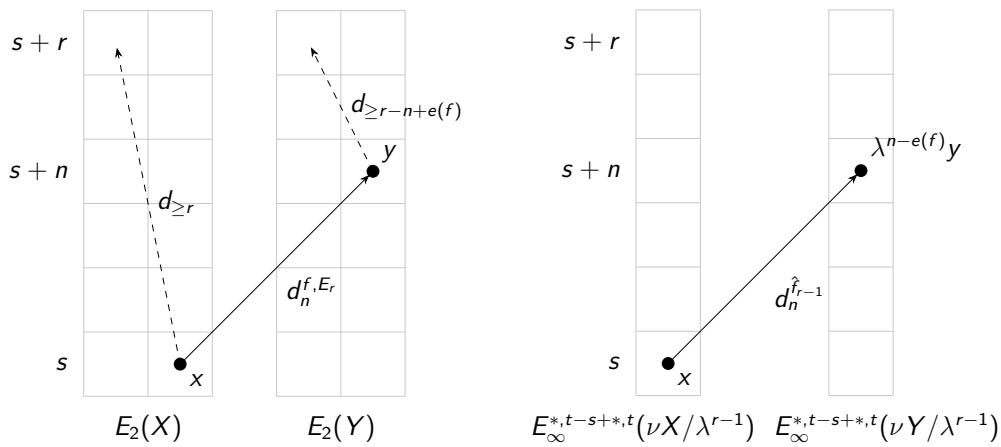

Figure 1: (f, E_r)-extension

Definition 4. *Let $x \in Z_{r-1}^{s,t}(X)$ and $y \in Z_{r-1-n+e(f)}^{s+n,t+n}(Y)$ for some*

$$ e(f) \leq n \leq r-2+e(f). $$

We say that there is an (f, E_r)-extension of level l from x to y, denoted by

$$ d_n^{f,E_r,l}(x) = y \tag{5} $$

if there exists a synthetic $\hat{f}_{r-1}$-extension

$$d_n^{\hat{f}_{r-1}}(\lambda^l x) = \lambda^{n-e(f)+l} y. \tag{6}$$

where $\lambda^l x$ is viewed as an element of the subgroup (2) with certain k, and $\lambda^{n-e(f)+l} y$ is viewed as an element of the quotient group (3).

We say that this (f, E_r)-extension of level l in (5) is essential *if the corresponding synthetic $\hat{f}_{r-1}$-extension in (6) is an essential differential in the $\hat{f}_{r-1}$-ESS.*

For $r = \infty$, we similar define an (f, E_∞)-extension of level l using the corresponding synthetic $\hat{f}$-extension.

Remark 7. *Consider an (f, E_r)-extension $d_n^{f,E_r,l}(x) = y$. The element y should be interpreted as a coset with indeterminacy given by boundaries in the Adams spectral sequence, plus the sum of images of $d_{<n}^{\hat{f}_{r-1}}$ differentials. This (f, E_r)-extension is* essential *if this coset does not contain 0.*

1.3 comparison at different levels

1.3.0.1

For $l < l'$, an extension of level l implies an extension of the same type of level l', but the latter might be inessential.

Proposition 8. *Suppose we have a (f, E_r)-extension*

$$d_n^{f,E_r,l'}(x) = y$$

Then this implies that either

1. *there is an essential $d_m^{f,E_r,l}(x) = z$ for some $m < n$*

2. *we have* $d_n^{f,E_r,l}(x) = y + z$ *for some* z *lying in the indeterminacy of* $d_n^{f,E_r,l'}$

1.4 crossings

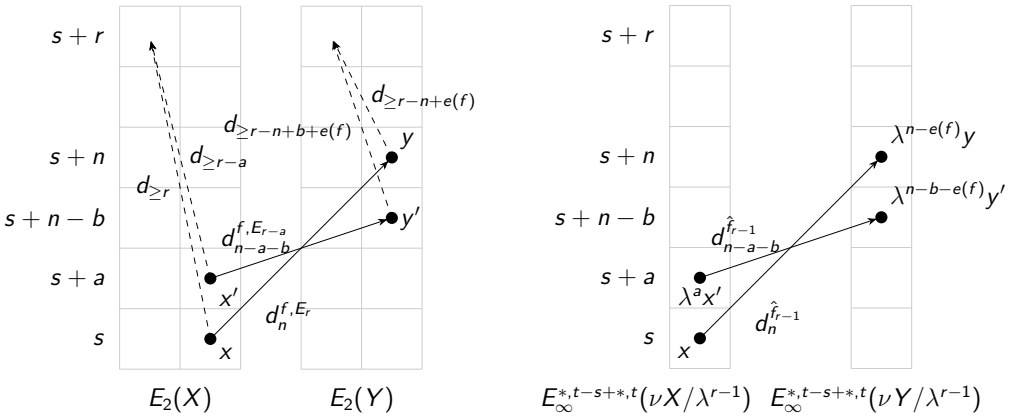

Figure 2: A crossing of an (f, E_r)-extension

1.4.0.1

The above Definition **??**, and Proposition **??** can be similarly defined and stated for the case $r = \infty$.

please make the precise statement.

1.4.0.2

A crossing of the (f, E_r)-extension of level l, $d_n^{f,E_r,l}(x) = y$, is defined as an essential (f, E_{r-a})-extension of level $l + a$ from some $x' \in Z_{r-1-a}^{s+a,t+a}(X)$ to

$$y' \in Z_{r-1-n+b+e(f)}^{s+n-b,t+n-b}(Y) \backslash B_{1+n-b-e(f)}^{s+n-b,t+n-b}(Y)$$

for $0 < a \leq r-2$ and $0 \leq b \leq n-a-e(f)$.

Proposition 9. *An* (f, E_r)*-extension of level* l, $d_n^{f,E_r,l}(x) = y$ *(5), has a crossing if and only if the synthetic* $\hat{f}_{r-1}$*-extension*

$$d_n^{\hat{f}_{r-1}}(\lambda^l x) = \lambda^{n-e(f)+l} y$$

has a crossing. Here a crossing of the latter has the form

$$d_{n-a-b}^{\hat{f}_{r-1}}(\lambda^{a+l}x') = \lambda^{n-b-e(f)+l}y', \tag{10}$$

which corresponds to the (f, E_{r-a})*-extension of level* l *from* x' *to* y'*.*

Proof *a crossing of the synthetic* $\hat{f}_{r-1}$*-extension* $d_n^{\hat{f}_{r-1}}(\lambda^l x) = \lambda^{n-e(f)+l}y$ *has the form (10).* □

Proposition 11. *If an* (f, E_r) *extension of level* l *has a crossing, then it is crossed by some* (f, E_r) *exteension of level* 0*, meanning that there is an essential extension at level* 0*, such that the implied extension at level* $a + l$ *satisies the conditions on degrees of a crossing.*

Proof *If there is a crossing, we use the alternatives in proposition 8.* □

Proposition 12. *If an extension at level* l *has no crossing, then the induced extension at level* $l' > l$ *also has no crossing.*

1.5 extension at E_r page and hidden extensions in *Adams* spectral sequences

Proposition 13. *Consider* $f : X \to Y$*,* $x \in Z_\infty^{s,t}(X)$ *and* $y \in Z_\infty^{s+n,t+n}(Y)$*. Then*

$$d_n^{f,E_\infty,l}(x) = y$$

implies that

$$d_n^f(x + B_\infty^{s,t}(X)) = y + B_\infty^{s,t}(Y).$$

Proof *we invert* λ*, then the synthetic homotopy groups is isomorphic to the classical stable homotopyp groups preserving the Adams filtration, so there is a morphsim of ESS.* □

2 The Generalized Leibniz Rule and Generalized Mahowald Trick

3 generalization Leibnitz rule

Now, we introduce the theorem of the Generalized Leibniz Rule, a valuable tool for computing new Adams differentials.

Theorem 14 (Generalized Leibniz Rule)**.** *Let $f : X \to Y$ be a map between two classical spectra. Suppose that $2 \leq n \leq r$, $e(f) \leq m \leq n-2+e(f)$, $l \geq e(f)$, and we have*

$$x \in Z_{r-1}^{s,t}(X), \quad y \in Z_{r-1-m+e(f)}^{s+m,t+m}(Y)$$

$$x_\infty \in Z_\infty^{s+r,t+r-1}(X), \quad y_\infty \in Z_\infty^{s+r+l,t+r+l-1}(Y)$$

and the following conditions hold:

1. $d_r(x) = x_\infty$,
2. $d_m^{f,E_n,k}(x) = y$,
3. $d_{r+l-m}(y) = y_\infty$
4. *the differential in (1) has no crossing on the E_n-page or (2) has no crossing.*
5. *the differential in (3) has no crossing.*

Then we have an Adams differential $d_l^{f,E_\infty,r-n+k}(x_\infty) = y_\infty$,

Proof *Consider the commutative diagram of synthetic spectra*

$$\begin{tikzcd}
\Sigma^{0,e(f)}\nu X/\lambda^{n-1} \arrow[r, "\hat{f}_{n-1}"] \arrow[d, "\delta_X"'] & \nu Y/\lambda^{n-1} \arrow[d, "\delta_Y"'] \\
\Sigma^{1,-n+1+e(f)}\nu X \arrow[r, "\hat{f}"'] & \Sigma^{1,-n+1}\nu Y.
\end{tikzcd}$$

By condition (1) and Proposition **??**, *we have*

$$d_r^{\delta_X}(\lambda^k x) = \lambda^{r-n+k} x_\infty. \tag{15}$$

By conditions (2) and (3), and Definition **??**, *we have*

$$d_m^{\hat{f}_{n-1}}(\lambda^k x) = \lambda^{m-e(f)+k} y \tag{16}$$

and

$$d_{r+l-m}^{\delta_Y}(y) = \lambda^{r+l-m-n} y_\infty$$

which implies

$$d_{r+l-m}^{\delta_Y}(\lambda^{m-e(f)+k} y) = \lambda^{r+l-n-e(f)+k} y_\infty$$

Applying Proposition **??** *and Proposition* **??** *to conditions (4) and (5), we know that the differential in (19) or (20) has no crossing, and that the differential in (21) has no crossing.*

Therefore, we can apply Corollary **??** *and obtain*

$$d_l^{\hat{f}}(\lambda^{r-n+k} x_\infty) = \lambda^{r+l-n-e(f)+k} y_\infty$$

□

Remark 17. *We can further generalize Theorem* **??** *by using the conditions in Theorem* **??** *rather than those Corollary* **??**. *We leave this generalization to the reader.*

try to do this.

Theorem 18 (Generalized Leibniz Rule)**.** *Let $f : X \to Y$ be a map between two classical spectra. Suppose that $2 \le n \le r$, $e(f) \le m \le n-2+e(f)$, $l \ge e(f)$, and we have*

$$x \in Z_{r-1}^{s,t}(X), \quad y \in Z_{r-1-m+e(f)}^{s+m,t+m}(Y)$$

$$x_\infty \in Z_\infty^{s+r,t+r-1}(X), \quad y_\infty \in Z_\infty^{s+r+l,t+r+l-1}(Y)$$

and the following conditions hold:

1. $d_r(x) = x_\infty$,
2. $d_m^{f,E_n,k}(x) = y$,
3. $d_l^{f,E_\infty,r-n+k}(x_\infty) = y_\infty$,
4. *the differential in (1) has no crossing on the E_n-page or (2) has no crossing.*
5. *the differential in (3) has no crossing.*

Then we have an Adams differential

$$d_{r+l-m}(y) = y_\infty.$$

if $m - e(f) + k < r + l - m$

Proof *Consider the commutative diagram of synthetic spectra*

$$\begin{tikzcd}
\Sigma^{0,e(f)}\nu X/\lambda^{n-1} \arrow[r, "\hat{f}_{n-1}"] \arrow[d, "\delta_X"'] & \nu Y/\lambda^{n-1} \arrow[d, "\delta_Y"'] \\
\Sigma^{1,-n+1+e(f)}\nu X \arrow[r, "\hat{f}"'] & \Sigma^{1,-n+1}\nu Y.
\end{tikzcd}$$

By condition (1) and Proposition **??**, *we have*

$$d_r^{\delta_X}(x) = \lambda^{r-n} x_\infty. \tag{19}$$

By conditions (2) and (3), and Definition **??**, *we have*

$$d_m^{\hat{f}_{n-1}}(\lambda^k x) = \lambda^{m-e(f)+k} y \tag{20}$$

and

$$d_l^{\hat{f}}(\lambda^{r-n+k}x_\infty) = \lambda^{r+l-n-e(f)+k}y_\infty. \tag{21}$$

Applying Proposition **??** *and Proposition* **??** *to conditions (4) and (5), we know that the differential in (19) or (20) has no crossing, and that the differential in (21) has no crossing.*

Therefore, we can apply Corollary **??** *and obtain*

$$d_{r+l-m}^{\delta_Y}(\lambda^{m-e(f)+k}y) = \lambda^{r+l-n-e(f)+k}y_\infty.$$

By Remark **??**, *this implies that* $d_{r+l-m}(y) = y_\infty$ *if* $m - e(f) + k < r + l - m$ □

3.1 Generalized Mahowald trick

Next, we discuss the Generalized Mahowald Trick.

In order to apply the Generalized Leibniz Rule, we need to provide a method for computing extensions on specific Adams E_k-pages, such as the (f, E_3)-extension $d_2^{f,E_3}(h_0h_4^2) = h_0p$ in Example **??**. This is provided by Theorem 23 (the Generalized Mahowald Trick). The crux of the proof of the Generalized Mahowald Trick lies in the following lemma.

Lemma 22 (May [**?**]). *Let* $X \to Y \to Z \to \Sigma X$ *and* $X' \to Y' \to Z' \to \Sigma X'$ *be distinguished triangles of (synthetic) spectra. By smashing these distinguished triangles together, we obtain the following commutative diagram of cofiber sequences:*

$$\begin{array}{ccccc}
X \wedge X' & \longrightarrow & Y \wedge X' & \longrightarrow & Z \wedge X' \\
\downarrow & & \downarrow & & \downarrow \\
X \wedge Y' & \longrightarrow & Y \wedge Y' & \longrightarrow & Z \wedge Y' \\
\downarrow & & \downarrow & & \downarrow \\
X \wedge Z' & \longrightarrow & Y \wedge Z' & \longrightarrow & Z \wedge Z'
\end{array}$$

If $a \in \pi_n(X \wedge Z')$ and $b \in \pi_n(Y \wedge Y')$ map to the same element in $\pi_n(Y \wedge Z')$, then there exists $c \in \pi_n(Z \wedge X')$ such that

1. *b and c map to the same element in $\pi_n(Z \wedge Y')$, and*
2. *a and c map to the same element via boundary maps in $\pi_{n-1}(X \wedge X')$.*

Proof *The proof follows directly from [**?**, Lemma 4.6].*

*In fact, consider V in Axiom TC3 of [**?**, Section 4] and the corresponding commutative diagram. By [**?**, Lemma 4.6] we know that V is the pull back of the following diagram.*

$$\begin{tikzcd} V \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & Y \wedge Y' \arrow[d] \\ X \wedge Z' \arrow[r] & Y \wedge Z' \end{tikzcd}$$

*Since a and b map to the same element in $\pi_n(Y \wedge Z')$, we know that we can find $v \in V$ such that v maps to a in $\pi_n(X \wedge Z')$ and b in $\pi_n(Y \wedge Y')$. Then we let $c = j_3(v) \in \pi_n(Z \wedge X')$, where $j_3 : V \to Z \wedge X'$ is the map in the commutative diagram in Axiom TC3 of [**?**, Section 4]. This lemma follows from the commutativity of the diagram.* □

Theorem 23 (Generalized Mahowald Trick)**.** *Consider a distinguished triangle of spectra*

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} \Sigma X$$

with $e(f) + e(g) + e(h) = 1$. Suppose that $r = n + m + l$, $n_1 = n - e(f) \geq 1$, $m_1 = m - e(g) \geq 0$, $l_1 = l - e(h) \geq 0$,

and

$$x \in Z_{n_1}^{s+l,t+l-1}(X),\ y \in Z_{m_1+1}^{s+n+l,t+n+l-1}(Y),$$
$$\bar{x} \in Z_{r-1}^{s,t}(Z), \qquad \bar{y} \in Z_{\infty}^{s+r,t+r-1}(Z),$$

such that

1. $d_l^{h,E_{r'},k}\bar{x} = x$, *where* $r' = r - m_1 = n_1 + l_1 + 1$,
2. $d_r\bar{x} = \bar{y}$,
3. *the* $(h, E_{r'})$*-extension in (1) has no crossing, or the Adams differential (2) has no crossing on the* $E_{r'}$*-page,*
4. $d_m^{g,E_{m_1+2},k}y = \bar{y}$.

Then we have $x \in Z_{n+m+e(h)}^{s+l,t+l-1}(X)$ *and*

$$d_n^{f,E_{n+m+1+e(h)},l_1}x = y$$

(See Figure 3.)

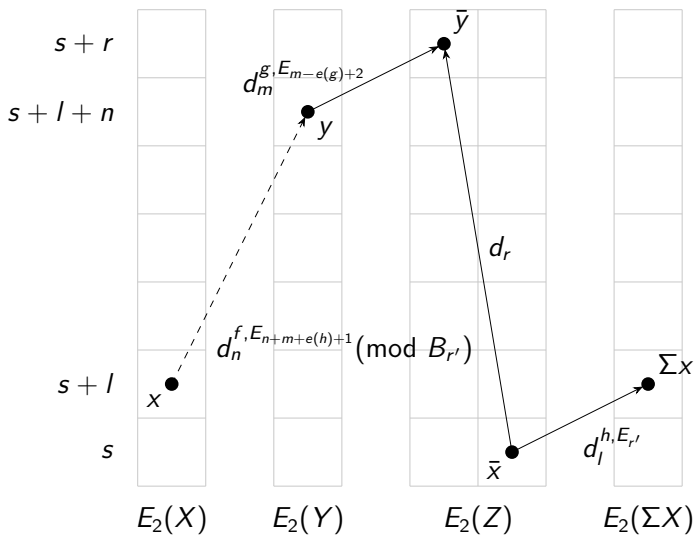

Figure 3: A demonstration of Theorem 23

Proof *Consider the following two distinguished triangles of*

$$\wedge \qquad \nu X \xrightarrow{\hat{f}} \Sigma^{0,-e(f)}\nu Y \xrightarrow{\hat{g}} \Sigma^{0,-e(f)-e(g)}\nu Z \xrightarrow{\hat{h}} \Sigma^{1,0}\nu X$$

$$\begin{tikzcd}
S^{0,0}/\lambda^r \arrow[d, "\rho"'] & & & [\lambda^{l_1}x] \arrow[d] \\
S^{0,0}/\lambda^{r'-1} \arrow[d, "\delta_{r'-1}"'] & & [\bar{x}] \arrow[r] \arrow[d] & [\lambda^{l_1}x] \\
S^{1,-r'+1}/\lambda^{m_1+1} \arrow[d, "\lambda^{r'-1}"'] & [y] \arrow[r] \arrow[d] & [\lambda^{m_1}\bar{y}] & \\
S^{1,0}/\lambda^r & [\lambda^{l_1}x] \arrow[r] & [\lambda^{r'-1}y] &
\end{tikzcd}$$

Figure 4: Elements in the homotopy groups of the smash products

synthetic spectra

$$\nu X \xrightarrow{\hat{f}} \Sigma^{0,-e(f)}\nu Y \xrightarrow{\hat{g}} \Sigma^{0,-e(f)-e(g)}\nu Z \xrightarrow{\hat{h}} \Sigma^{0,-1}\nu\Sigma X = \Sigma$$

$$S^{0,0}/\lambda^r \xrightarrow{\rho} S^{0,0}/\lambda^{r'-1} \xrightarrow{\delta_{r'-1}} S^{1,-r'+1}/\lambda^{m_1+1} \xrightarrow{\lambda^{r'-1}} S^{1,0}/\lambda$$

and their smash product. See Figure 4.

By condition (1), we have

$$d_l^{\hat{h}_{r'-1}}(\lambda^k\bar{x}) = \lambda^{l_1+k}x \tag{24}$$

where $\hat{h}_{r'-1}$ *is the map* $\Sigma^{0,e(h)}\nu Z/\lambda^{r'-1} \to \nu X/\lambda^{r'-1}$ *induced by* $\hat{h}$*. By condition (2), we have*

$$d_r^{\delta_{r'-1}}(\bar{x}) = \lambda^{m_1}\bar{y}. \tag{25}$$

Applying Proposition **??** *and Proposition* **??** *to condition (3), we know that the differential in (24) or the differential in (25) has no crossing. Hence, there exists*

$$[\lambda^k\bar{x}] \in \{\lambda^k\bar{x}\} \subset \pi_{t-s,t}(\nu Z/\lambda^{r'-1})$$

such that

$$\hat{h}_{r'-1}([\lambda^k\bar{x}]) \in \{\lambda^{l_1+k}x\} \subset \pi_{t-s-1,t-1+e(h)}(\nu X/\lambda^{r'-1})$$

and

$$\delta_{r'-1}([\lambda^k\bar{x}]) \in \{\lambda^{m_1+k}\bar{y}\} \subset \pi_{t-s-1,t+r'-1}(\nu Z/\lambda^{m_1+k_1+1}).$$

By condition (4), we have

$$d_m^{\hat{g}_{m_1+1}}(\lambda^k y) = \lambda^{m_1+k}\bar{y}.$$

This implies that there exists

$$[\lambda^k y] \in \{\lambda^k y\} \subset \pi_{t-s-1,t+n+l-1}(\nu Y/\lambda^{m_1+k+1})$$

such that

$$\hat{g}_{m_1+k+1}([\lambda^k y]) \in \{\lambda^{m_1+k}\bar{y}\} \subset \pi_{t-s-1,t+r'-1}(\nu Z/\lambda^{m_1+k+}$$

where $\hat{g}_{m_1+k+1}$ *is the map* $\Sigma^{0,e(g)}\nu Y/\lambda^{m_1+k+1} \to \nu Z/\lambda^{m_1+k+1}$ *induced by* $\hat{g}$.

Due to degree reasons (Proposition **??***),* $\lambda^{m_1+k}\bar{y}$ *detects a unique element in homotopy,*

$$[\lambda^{m_1+k}\bar{y}] \in \pi_{t-s-1,t+r'-1}(\nu Z)/\lambda^{m_1+k+1}.$$

Therefore, we have

$$\delta_{r'-1}([\lambda^k\bar{x}]) = \hat{g}_{m_1+k+1}([\lambda^k y]) = [\lambda^{m_1+k}\bar{y}],$$

By Lemma 22, there exists

$$[\lambda^{l_1+k}x] \in \pi_{t-s-1,t-1+e(h)}(\nu X/\lambda^r),$$

such that

$$\rho([\lambda^{l_1+k}x]) = \hat{h}_{r'-1}([\lambda^k\bar{x}]), \tag{26}$$

$$\hat{f}_r([\lambda^{l_1+k}x]) = \lambda^{r'-1}[\lambda^k y], \tag{27}$$

where ρ *is the map* $\nu X/\lambda^r \to \nu X/\lambda^{r'-1}$*, and* $\hat{f}_r$ *is the map* $\Sigma^{0,e(f)}\nu X/\lambda^r \to \nu Y/\lambda^r$ *induced by* $\hat{f}$.

The equality in (26) indicates that $[\lambda^{l_1+k}x]$ *can be lifted to the* E_∞*-page of* $\nu X/\lambda^r$*, so we have*

$$\lambda^k x \in Z^{s+l,t+l-1}_{n+m+e(h)}(X).$$

The equation in (27) indicates that

$$d_n^{\hat{f}_r}(\lambda^{l_1+k}x) = \lambda^{r'-1+k}y.$$

By definition, this is equivalent to

$$d_n^{f,E_{n+m+1+e(h)},l_1+k}x = y$$

□

3.1.0.1

There is also a version of the generalized Mahowald trick where the assumtion is g and h extensions at level k, try to do this generalization.

3.2 Extensions in different pages

We refer to Theorem 23 as the Generalized Mahowald Trick, as this approach was first utilized by Mahowald and his collaborators in various works (see, for example, [**?**]), particularly in the case where $m_1 = l_1 = 0$. The synthetic setting advances this method by allowing for the consideration of cases where $m_1 > 0, l_1 > 0$ as well. The outcome of the Generalized Mahowald Trick, as stated in Theorem 23, is an f-extension on a specific Adams page, such as the (f, E_3)-extension $d_2^{f,E_3}(h_0h_4^2) = h_0p$ in Example **??**(2) and Example **??**. In practice, the source of an (f, E_r)-extension may survive to later Adams pages, prompting interest in the $(f, E_{>r})$-

extensions, such as the (f, E_∞)-extension $d_2^{f,E_\infty}(h_0h_4^2) = h_0p$ in Example **??**(3), which is applied in Example **??**. The following Propositions 28 and 31 describe the relationships between extensions across different pages.

Proposition 28. *Suppose that we have an (f, E_r)-extension $d_n^{f,E_r}(x) = y$, where $x \in Z_{r-1}^{s,t}(X)$ and $y \in Z_{r-1-n+e(f)}^{s+n,t+n}(Y)$. Then for all $2 \le r' < r$, we also have*

$$d_n^{f,E_{r'}}(x) = y. \tag{29}$$

Furthermore, if $d_n^{f,E_r}(x) = y$ is essential and $n \le r' - 2 + e(f)$, then (29) is inessential if and only if there exists some $0 < a' \le n - e(f)$ and an element

$$x' \in Z_{r'-1-a'}^{s+a',t+a'}(X) \backslash Z_{r-1-a'}^{s+a',t+a'}(X)$$

such that

$$d_{n-a'-b}^{\hat{f}_{r'-1}}(\lambda^{a'}x') = \lambda^{n-b-e(f)}y'$$

for some $b \ge 0$.

Proof *Consider the following commutative diagram:*

$$\begin{array}{ccc} \Sigma^{0,e(f)}\nu X/\lambda^{r-1} & \xrightarrow{\hat{f}_{r-1}} & \nu Y/\lambda^{r-1} \\ \rho_X\downarrow & & \rho_Y\downarrow \\ \Sigma^{0,e(f)}\nu X/\lambda^{r'-1} & \xrightarrow{\hat{f}_{r'-1}} & \nu Y/\lambda^{r'-1} \end{array}$$

*By Corollary **??**, (ρ_X, ρ_Y) induces a map from the $\hat{f}_{r-1}$-ESS to the $\hat{f}_{r'-1}$-ESS. Therefore, by naturality,*

$$d_n^{\hat{f}_{r-1}}(x) = \lambda^{n-e(f)}y \text{ implies } d_n^{\hat{f}_{r'-1}}(x) = \lambda^{n-e(f)}y,$$

which, by definition, is

$$d_n^{f,E_{r'}}(x) = y.$$

Next, we prove the second part of the proposition. By Definition **??**, *the* (f, E_r)*-extension (29) is inessential if and only if there exists* $0 < a \leq n - e(f)$ *and*

$$\lambda^a x' \in E_\infty^{s+a,t+a,t}(\nu X/\lambda^{r'-1}) \cong Z_{r'-1-a}^{s+a,t+a}(X)/B_{1+a}^{s+a,t+a}(X)$$

such that

$$d_{n-a}^{\hat{f}_{r'-1}}(\lambda^a x') = \lambda^{n-e(f)} y, \tag{30}$$

and this differential is not induced by (ρ_X, ρ_Y)*, as we assume that* $d_n^{f,E_r}(x) = y$ *is essential.*

There are two scenarios where this differential is not induced by (ρ_X, ρ_Y)*. The first case occurs when* $\lambda^a x'$ *is not in the image of* ρ_X *at all, which is equivalent to* $x' \notin Z_{r-1-a}^{s+a,t+a}(X)$*. (See Figure 5.)*

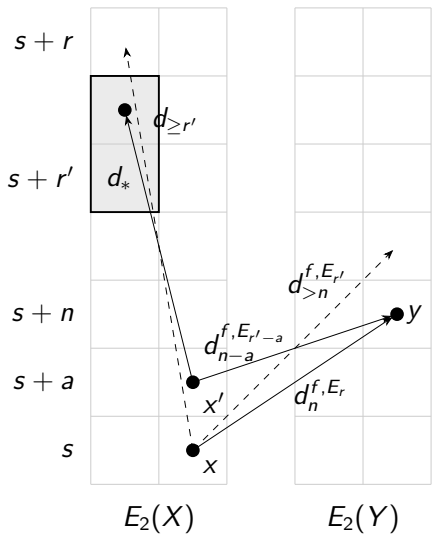

Figure 5: Extensions across pages

The second case occurs when $\lambda^a x'$ *is in the image of* ρ_X*, but it supports an essential* $\hat{f}_{r-1}$*-extension that is strictly shorter than the differential (30).*

To further explore this scenario, we replace x with $\lambda^a x'$ and analyze successive essential extensions. This process is repeated iteratively until the first case is reached. Ultimately, this leads to the existence of some $0 < a' \leq n - e(f)$ and an element

$$x'' \in Z^{s+a',t+a'}_{r'-1-a'}(X) \backslash Z^{s+a',t+a'}_{r-1-a'}(X)$$

such that

$$d^{\hat{f}_{r'-1}}_{n-a'-b}(\lambda^{a'} x'') = \lambda^{n-b-e(f)} y'$$

for some $b > 0$. This represents a crossing of $d_n^{\hat{f}_{r'-1}}(x) = \lambda^{n-e(f)} y$.

□

The following Corollary 31 is the contrapositive statement of the second part of Proposition 28.

Corollary 31. *Suppose $r \geq r'$ and there exists an $(f, E_{r'})$-extension*

$$d_n^{f,E_{r'}}(x) = y$$

for $x \in Z^{s,t}_{r-1}(X)$ and $y \in Z^{s+n,t+n}_{r'-1-n+e(f)}(Y)$. Assume that this extension has no crossing of the form $d^{f,E_{r'-a}}_{n-a-b}(x') = y'$ for any $b > 0$, $0 < a \leq n - e(f)$, and

$$x' \in Z^{s+a,t+a}_{r'-1-a}(X) \backslash Z^{s+a,t+a}_{r-1-a}(X).$$

Under these conditions, we also have an (f, E_r)-extension

$$d_n^{f,E_r}(x) = y.$$